

5	(a) work done in moving unit positive charge from infinity (to the point)	M1 A1	[2]
	(b) (i) inside the sphere, the potential would be constant	B1	[1]
	(ii) for point charge, V_x is constant co-ordinates clear and determines two values of V_x at least 4 cm apart conclusion made clear	B1 M1 A1	[3]
	(c) $q = 4\pi\epsilon_0 Vx$ $q = 4\pi \times 8.85 \times 10^{-12} \times 180 \times 1.0 \times 10^{-2}$ $= 2.0 \times 10^{-10} \text{ C}$	M1 A1	[2]